- Rebuilt from my own research project, Play to Learn in VR (P01.01), continuing the student study from the Project Prospectus, Definitions Practice, and Codebook examples: one coder, 150 tweets from 2021 about playing in VR.
- Everything under "The example" is what a student would submit: one document with the plan and the pilot, plus the coded pilot data. The draw, the codes, and the agreement figures are real: the tweets were drawn by the script shown, coded twice with the codebook, and the figures are the script's own output. The minutes and the dates are a student's schedule.
- No tweet is quoted. Each case is paraphrased, and the data file carries tweet IDs and codes, not text, so a search cannot lead back to an account.
- The rubric map at the end shows which grading row each part earns and why.

The example

Sampling Plan and Pilot: Learning Talk and VR Tone in Tweets About Playing in VR

Data file: MC451-pilot-codes.csv (20 rows, one per tweet).

Part 1: Sampling plan

Population

My claim covers English-language tweets, not retweets, posted from January 1 through December 31, 2021, that mention virtual reality (VR) together with the word play. Its edges: Twitter only; original posts, replies, and quote tweets, but not retweets; any account, whether a player, a company, or a school; and play in any sense.

Sampling frame

I draw from the Play to Learn in VR analysis file, which holds the text of 6,108 tweets from a 2021 archive of 100,363 collected through the Twitter Academic Research API. The file keeps every one of the 3,054 archive tweets that contain one of 34 learning words, plus 3,054 drawn at random from the other 97,309.

The frame differs from the population in three ways. Tweets the archive's search missed, or that were deleted before collection, are not in it. Learning-word tweets are half of the frame but 3 percent of the archive, so I report each stratum separately rather than one percentage across all 150. And its text has digits and punctuation stripped (decision rule 3).

Method

I use a stratified random sample: the frame is split into two strata, tweets with a learning word and tweets without one, and 75 are drawn at random from each, with the seed written in the script. H1 compares tweets that refer to learning with those that do not, and a simple random draw from the archive would give about five learning-word tweets in 150, too few to compare.

Size, with the arithmetic

I will code 150 tweets. I timed the pilot: 20 tweets in 20 minutes, 60 seconds per tweet with four variables. The full pass is 150 times 60 seconds, 9,000 seconds, or 2.5 hours. Building the codebook and running this pilot took about 1 hour, so coding totals about 3.5 hours, inside the two-to-four-hour budget, with roughly the first third spent building. I used the sizing table's row for short posts (20 to 60 words, 75 seconds, 150 items); my rate is faster than the row's, so 150 fits.

Date range and filters

Applied in this order in R:

1. Posted from January 1 through December 31, 2021, by the tweet's own timestamp.
2. Language tag English.
3. Not flagged as possibly sensitive by Twitter.
4. Of tweets with identical text (a bot reposting one promotion), only the lowest tweet ID is kept.
5. The 30 immersion tweets that built the codebook are removed.

Left out: retweets (excluded at collection), non-English, sensitive-flagged, and repeated tweets, and the immersion tweets. Draws 76 to 90 in each stratum are a reserve list, used in order to replace any tweet marked out of frame.

Step	Tweets left	Removed
Frame: VR and play tweets with text	6,108	
Posted 01-01 through 12-31, 2021	6,108	0
Language tag English	6,105	3
Not flagged sensitive	5,833	272
First copy of repeated text only	5,236	597
Immersion tweets removed	5,206	30
Learning-word stratum / other stratum	2,359 / 2,847	
Drawn: sample / reserve, per stratum	75 / 15	

Part 2: Pilot

Items coded

I coded the first 10 draws from each stratum, 20 tweets, with the codebook frozen on 10-30. MC451-pilot-codes.csv has one row per tweet: ID, stratum, draw order, date, my four codes from each pass, and the like count, looked up after coding.

Stability check

I coded all 20 tweets on 10-31 and recoded all 20 on 11-03, 72 hours later. For the second pass I copied the IDs and text into a fresh sheet with no code columns, in a shuffled order, so my first codes were out of sight. The passes agreed on 77 of 80 decisions; alpha is from v2v::reliability(), with the code below.

Variable	Passes agree	Krippendorff's alpha
Learning reference	20 of 20 (100%)	1.00
Tone toward VR	19 of 20 (95%)	.90
Affordance named	19 of 20 (95%)	.91
Constraint named	19 of 20 (95%)	.86

Each disagreement was a tweet no rule covered, and each produced a revision (edge cases 4 to 6). These figures measure my own consistency, not agreement between coders.

Edge cases

1. **VR that plays a role.** Two tweets used play in another sense (VR playing a part). Decision: in frame, coded as usual. Rule: decision rule 1, revision A.
2. **Learning that VR prompts but that happens elsewhere.** A sword game sent a player to practice a real sword art; a teacher planned a body-awareness lesson without VR. Decision: learning reference 0. Rule: learning reference rule 6, revision B.
3. **A lesson learned, a problem implied.** A player said they had learned not to play VR right after waking up. Decision: learning reference 0, an idiom; constraint 0, since the problem is hinted at, not named. Rules: learning reference rule 1 and constraint rule 5, revisions C and F.
4. **A harm, not a hardware problem.** A writer blamed VR, with gaming and film, for cutting people off from the world. My passes split, 0 then 4. Decision: 4. Rule: constraint rule 6, revision F.
5. **A product's claim about itself.** A queue-simulator promotion called its new levels bug-free. Tone split, 3 then 2. Decision: 2; bug-free describes the product, not the experience. Rule: tone rule 2, revision D.

6. **Access for a named group.** An announcement said VR gave students with disabilities a safe, immersive experience. Affordance split, 5 then 1. Decision: 5; the benefit is who gets to take part. Rule: affordance category 5 and rule 5, revision E.
7. **A foreign-language hashtag string.** Game names and a German contest word, tagged English. Decision: decision rule 4 for hashtag strings settled it. No new rule.
8. **Four benefits at once.** A cryptocurrency metaverse promotion offered education, earnings, games, and social networks. Decision: the existing tie rule picked the longest, earnings (5). No new rule.

Revisions the pilot forced

Revision	Before	After
A. Decision rule 1	Before coding, check that the tweet is about virtual reality. If VR appears only inside another word, a username, or an unrelated abbreviation, and the tweet is about something else, mark it out of frame, code nothing, and replace it with the next tweet on the reserve list.	The same rule, plus: The word play may carry any sense. A tweet saying VR will play a role in training is in frame and is coded like any other.
B. Learning reference	(No rule.)	6. Learning that VR prompts but that happens outside VR (a sword game that sends a player to a real sword class, a lesson planned without VR) is coded 0.
C. Learning reference rule 1	A learning word inside a game, product, or film title, or inside an idiom (learned my lesson, hype train), is coded 0 unless another clause describes learning.	A learning word inside a game, product, or film title, or inside an idiom (learned my lesson, today I learned, hype train), is coded 0 unless another clause describes learning.
D. Tone rule 2	An advertisement that lists features or a release date with no evaluative word is coded 2.	An advertisement that lists features, a release date, or a quality claim about the product itself (bug-free, new levels, sharper graphics) is coded 2. Only a word that judges the experience (amazing, fun, the best) makes it 3.

Revision	Before	After
E. Affordance named	5 = other benefit (exercise, wellbeing, anything else). (No rule 5.)	5 = other benefit (exercise, wellbeing, access for people who could not otherwise take part, anything else). Rule 5: When VR is said to open an experience to a named group who would otherwise miss it, code 5, even if the experience is also called immersive.
F. Constraint named	(No rules 5 or 6.)	5. A problem the tweet hints at without naming is coded 0. 6. A harm the writer says VR does to people (cutting them off from others, blurring fantasy and reality) is coded 4, because it gives a reason to use VR less.

The revised codebook is frozen before the full pass, each change logged with its date and the tweet behind it.

R code

The draw, from 01-draw-sample.R (the filters above come first in the same script):

```
set.seed(1106)
drawn <- frame %>%
  group_by(stratum) %>%
  slice_sample(n = 90) %>%
  mutate(draw_order = row_number()) %>%
  ungroup() %>%
  mutate(role = if_else(draw_order <= 75, "sample", "reserve"))
```

The stability figures, from 02-pilot-stability.R, with its output:

```
options(scipen = 999)
library(tidyverse)
library(v2v)

pilot <- read_csv(
  "MC451-pilot-codes.csv",
  col_types = cols(tweet_id = col_character())
)

pilot %>%
```

```
  summarize(
    items = n(),
    learning = mean(learning_p1 == learning_p2),
    tone = mean(tone_p1 == tone_p2),
    affordance = mean(affordance_p1 == affordance_p2),
    constraint = mean(constraint_p1 == constraint_p2)
  )
# A tibble: 1 x 5
#   items learning  tone affordance constraint
#   <int>    <dbl> <dbl>      <dbl>      <dbl>
# 1    20        1  0.95       0.95       0.95

reliability(pilot$tone_p1, pilot$tone_p2, method = "alpha")$statistic
# [1] 0.8992248
reliability(pilot$affordance_p1, pilot$affordance_p2, method =
"alpha")$statistic
# [1] 0.9111617
reliability(pilot$constraint_p1, pilot$constraint_p2, method =
"alpha")$statistic
# [1] 0.8602151
```

Rubric map

The plan rows total 35, the pilot data rows 15 (the stability check is graded inside them), and edge cases and revisions 25.

Rubric row	Points	Where the example earns it	Why it earns the full row
Plan: population	6	Population	It defines everything the claim covers and states every edge: the platform, which accounts (any), which posts (no retweets), the language, and the period.
Plan: sampling frame	7	Sampling frame	It names the actual file drawn from, with its size and how it was built, and says in three specific ways how it differs from the population instead of treating the two as one.
Plan: method	7	Method	It names the method (stratified random) and gives a reason tied to this question: H1 compares two groups, and one group is 3 percent of the archive.

Rubric row	Points	Where the example earns it	Why it earns the full row
Plan: size, with arithmetic	10	Size, with the arithmetic	It gives the number (150) and the arithmetic from a rate timed on the student's own pilot (60 seconds times 150, 2.5 hours, 3.5 hours with the build), inside the two-to-four-hour budget, and names the sizing-table row used.
Plan: date range and filters	5	The five numbered rules and the counts table	Dates, language, the sensitive flag, repeats, and the immersion set are stated as rules anyone could rerun, with what is left out and how many tweets each rule removed.
Pilot data: items coded	7	Items coded, and MC451-pilot-codes.csv	Twenty items, the top of the ten-to-twenty range, coded with the student's own codebook and submitted as a CSV with one row per item.
Pilot data: stability check	8	Stability check, its table, and the R code	All 20 items (more than the quarter required) recoded 72 hours later with the first codes hidden, agreement reported per variable, and the change each disagreement led to named.
Edge cases	10	The eight edge cases	Every case met in the pilot is listed with the decision and the rule it became, including the two that an existing rule already settled.
Revisions forced	15	The revisions table	Each of the six changes is shown with its exact wording before and after, not only described.

Length: about 880 words of the student's own writing across the plan and the pilot, not counting headings, tables, or code, inside the assignment's range, plus 20 coded items in the CSV.